Replication Guide

This document explains how to reproduce the experimental results.

Step 1: Project Structure

Ensure the following structure:

```
project/
│── dataset/
│   ├── processed_adult.csv
│   └── processed_compas.csv
│── DNN/
│   ├── model_processed_adult.h5
│   ├── scaler_adult.pkl
│   ├── model_processed_compas.h5
│   └── scaler_compas.pkl
│── train_model.py
│── experiments.py
│── results/
```

Step 2: Install Dependencies
pip install numpy pandas scikit-learn tensorflow matplotlib scipy joblib

Step 3: Train Models
If models are not provided:
python train_model.py

Step 4: Run Experiments
python experiments.py

This will:

- Load datasets and models
- Generate counterfactual test pairs
- Detect discrimination
- Run 7 experiments
- Save results automatically

Step 5: Output Files
Results are saved in:

results/

CSV files:

- exp1_adult_main.csv
- exp2_k_variation.csv
- exp3_sensitive_adult.csv
- exp4_budget_adult.csv
- exp5_compas_main.csv
- exp6_sensitive_compas.csv
- exp7_cross_dataset.csv

Figures:

- exp1_adult_main.png
- exp2_k_variation.png
- exp3_sensitive_adult.png
- exp4_budget_adult.png
- exp5_compas_main.png
- exp6_sensitive_compas.png
- exp7_cross_dataset.png

Step 6: Expected Results

Based on experiments:
- ADULT dataset improvement: ~35–40%
- COMPAS dataset improvement: ~90%
- Statistical significance: $p \approx 0.03$

The cluster-guided approach should:
- Consistently outperform random search
- Detect more discriminatory instances
- Perform robustly across configurations

Step 7: Reproducibility Details
- Random seed: 42
- Runs per experiment: 5
- Statistical test: Wilcoxon signed-rank

Notes
- Minor variations may occur
- Runtime: ~10–15 minutes
- No GPU required

Following these steps will reproduce all results presented in the report.